ModelIt!

Teacher's Guide

Is Climate Change Reversible?

Modeling Carbon Cycle Dynamics and Climate Tipping Points

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-ESS3-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-ESS3-5

Analyze geoscience data and the results from global climate models to make an evidence-based forecast of the current rate of global or regional climate change and associated future impacts to Earth systems.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models of the carbon cycle to predict how changes in emissions, absorption, and feedback loops affect global temperature trajectories over decades.
Disciplinary Core Idea	**ESS3.D Global Climate Change** — Human activities are responsible for the increase in greenhouse gases, and the resulting climate change has significant consequences for Earth systems including feedback mechanisms that can accelerate warming beyond initial forcing.
Crosscutting Concept	**Stability and Change** — Students analyze how the climate system can shift from a relatively stable state to a new equilibrium through positive feedback loops and tipping points, and evaluate the conditions under which stability can be maintained or restored.

CAST Testing Alignment

MULTIPLE CHOICE:

A student\'s climate model shows that even after reducing fossil fuel emissions to zero, Global Temperature Anomaly continues rising for 25 years before stabilizing. Which component of the model best explains this continued warming? A) Increased fossil fuel emissions B) Thermal inertia and activated feedback loops C) Decreased solar output D) Volcanic activity

CONSTRUCTED RESPONSE:

Using your carbon cycle model, explain why a 50% reduction in emissions is not expected to produce a 50% reduction in warming. Reference at least three components and one feedback loop from your model in your explanation.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The global carbon cycle is one of Earth\'s most fundamental biogeochemical processes, moving approximately 200 gigatons of carbon between the atmosphere, ocean, biosphere, and lithosphere each year. For millions of years, these flows were roughly balanced, maintaining atmospheric CO2 between 180 and 280 parts per million. Since the Industrial Revolution, human activities have added an additional 36 gigatons of CO2 per year, overwhelming the natural cycle and driving atmospheric concentrations past 424 ppm — the highest level in at least 800,000 years.

The Carbon Cycle: Natural Balance Disrupted

Earth\'s carbon cycle operates on multiple timescales. The fast carbon cycle involves photosynthesis, respiration, and ocean gas exchange, moving carbon between the atmosphere and biosphere on timescales of days to decades. The slow carbon cycle involves geological processes like weathering, sedimentation, and volcanism that operate over millions of years. Fossil fuels represent carbon that was removed from the fast cycle and stored geologically over hundreds of millions of years. By burning fossil fuels, humans are reintroducing this stored carbon into the fast cycle at a rate roughly 100 times faster than natural geological processes — essentially compressing millions of years of carbon release into a few centuries.

Feedback Loops: Why Warming Amplifies Itself

The climate system contains multiple positive feedback loops that amplify initial warming. The ice-albedo feedback is the most visually dramatic: as Arctic sea ice melts, the reflective white surface is replaced by dark ocean water that absorbs 94% of incoming solar radiation instead of reflecting 80% of it, causing additional warming. Permafrost feedback is potentially the most dangerous: Arctic permafrost contains an estimated 1,500 gigatons of carbon — roughly twice the total carbon in the atmosphere — and as it thaws, microbial decomposition releases CO2 and methane, a greenhouse gas 80 times more potent than CO2 over 20 years. Water vapor feedback amplifies warming by approximately 50%: warmer air holds more water vapor, which is itself a greenhouse gas.

Tipping Points: Thresholds of No Return

Climate tipping points are critical thresholds beyond which a system shifts irreversibly to a new state. Scientists have identified several potential tipping points: collapse of the West Antarctic Ice Sheet (committed to 3-5 meters of sea level rise once triggered), dieback of the Amazon rainforest (converting the world\'s largest carbon sink into a carbon source), shutdown of the Atlantic Meridional Overturning Circulation (dramatically cooling Europe while warming the tropics), and widespread permafrost collapse. Current warming of 1.2 degrees Celsius may have already committed some of these tipping points. Research published in Science in 2022 concluded that several tipping points become likely between 1.5 and 2.0 degrees of warming — the targets of the Paris Agreement.

Carbon Sinks and the Race Against Time

Natural carbon sinks currently remove about 50% of human CO2 emissions — the ocean absorbs roughly 25% and land ecosystems absorb roughly 25%. However, the efficiency of these sinks is declining as temperatures rise. Warmer oceans dissolve less CO2 (just as warm soda goes flat faster than cold soda). Stressed forests from drought, fire, and heat become carbon sources rather than sinks. Technological carbon capture exists but remains extremely expensive at $250-600 per ton of CO2 and currently removes less than 0.01% of annual emissions. Nature-based solutions like reforestation, wetland restoration, and regenerative agriculture offer more cost-effective carbon removal but require massive scale to meaningfully impact atmospheric concentrations.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Atmospheric CO2, Fossil Fuel Emissions, Ocean Carbon Absorption, Terrestrial Carbon Storage, Global Temperature Anomaly, Ice-Albedo Feedback, and Permafrost Methane Release as key components of the climate system.

E **Establish Relationships**
Students determine that fossil fuel emissions drive atmospheric CO2 increases, that temperature anomaly triggers self-reinforcing feedback loops, and that ocean and terrestrial sinks have limited and declining capacity to absorb excess carbon.

V **Visualize & Model**
Students build a computational model showing how carbon flows between reservoirs, how feedback loops amplify warming, and how tipping points create irreversible shifts in the climate system.

E **Evaluate Outcomes**
Students run business-as-usual, immediate net-zero, and enhanced sinks scenarios to test predictions about the reversibility of climate change under different intervention strategies.

R **Revise & Extend**
Students add aerosol masking, deep ocean circulation, or methane clathrate components to explore how additional processes affect climate trajectories and tipping point risks.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Today we\'re investigating the biggest question in environmental science: Is Climate Change Reversible? By the end of class, you\'ll have model-based evidence to answer this."*

Watch For

Show cover image. Quick poll: Thumbs up if you think climate change is reversible, thumbs down if not, sideways if unsure. Note the distribution.

Slide 2: Learning Objectives

3 min

Say: *"Here are our four objectives. Notice we\'re not just learning what climate change is — we\'re modeling the system dynamics that determine whether it can be reversed."*

Watch For

Have students read objectives aloud. Pre-teach vocabulary: carbon flux, positive feedback loop, tipping point, carbon sink. Use the warm soda analogy for ocean absorption.

Slide 3: Big Question

3 min

Say: *"If we stopped all emissions tomorrow, would the planet cool back down? This is not a simple yes-or-no question — it depends on feedback loops and tipping points we\'re about to model."*

Watch For

Quick-write: Students write their initial prediction and reasoning in 2-3 sentences. They\'ll revisit this after modeling.

Slide 4: LEVER Framework

2 min

Say: *"We\'ll use the LEVER framework to build a computational model of the carbon cycle. This model will let us test different emission scenarios and see which ones prevent tipping points."*

Watch For

Preview each LEVER step. Emphasize that modeling lets us test scenarios we can\'t experiment with in real life — we only have one Earth.

Slide 5: Activity 1: Components

8 min

Say: *"Which of these components are inputs from outside the climate system, and which are internal responses? Think about what drives the system versus what the system produces."*

Watch For

Guide sorting: External = Atmospheric CO2 (set by cumulative emissions), Fossil Fuel Emissions (human decision). Internal = the rest. Use think-pair-share for reasoning about borderline cases.

Slide 6: Activity 2: Connections

8 min

Say: *"When Fossil Fuel Emissions increase, what happens to Atmospheric CO2? When temperature rises, what happens to Ice-Albedo Feedback? Look for loops where effects cycle back to amplify the original cause."*

Watch For

Students draw arrows with + or - signs. Facilitate discovery of positive feedback loops. Ask: Where do you see a cycle that keeps reinforcing itself?

Slide 7: Activity 3: Simulation

10 min

Say: *"Now we test our model. Run all three scenarios and watch carefully — the business-as-usual scenario may shock you, and the net-zero scenario will challenge your assumptions."*

Watch For

Students run each scenario, record predictions versus observations. Key discussion: Why does temperature keep rising even after emissions hit zero? This is the critical insight.

Slide 8: Discoveries

5 min

Say: *"So is climate change reversible? Partially — and the answer depends on how quickly we act. Let\'s look at what our model revealed about feedback loops, tipping points, and the cost of delay."*

Watch For

Class discussion of discoveries. Have students compare their initial predictions to model evidence. Key question: What surprised you most?

Slide 9: STEM Challenge

12 min

Say: *"Now apply your model knowledge. Your city faces sea level rise, stronger storms, water scarcity, and heat waves. Some of these are already locked in. Design a plan that addresses both prevention and preparation."*

Watch For

Groups design 30-year climate resilience plans using model data. Must address both mitigation and adaptation. Present key strategies and justify with evidence from simulations.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Atmospheric CO2, Fossil Fuel Emissions
- **INTERNAL:** Ocean Carbon Absorption, Terrestrial Carbon Storage, Global Temperature Anomaly, Ice-Albedo Feedback, Permafrost Methane Release

Relationships

- **Fossil Fuel Emissions to Atmospheric CO2:** POSITIVE (+) — Burning fossil fuels releases stored carbon into the atmosphere, directly increasing CO2 concentration. Higher emission rates produce faster CO2 accumulation.
- **Atmospheric CO2 to Global Temperature Anomaly:** POSITIVE (+) — Higher atmospheric CO2 strengthens the greenhouse effect, trapping more outgoing infrared radiation and increasing global average surface temperature.
- **Global Temperature Anomaly to Ice-Albedo Feedback:** POSITIVE (+) — Higher temperatures melt reflective ice and snow, exposing darker surfaces that absorb more solar energy, which drives additional warming in a self-reinforcing loop.
- **Global Temperature Anomaly to Permafrost Methane Release:** POSITIVE (+) — Higher temperatures thaw Arctic permafrost, releasing stored methane and CO2 that further increase atmospheric greenhouse gas concentrations and drive additional warming.
- **Global Temperature Anomaly to Ocean Carbon Absorption:** NEGATIVE (-) — Warmer ocean water dissolves less CO2, reducing the ocean\'s capacity as a carbon sink. This means a larger fraction of emissions remains in the atmosphere, accelerating warming.
- **Terrestrial Carbon Storage to Atmospheric CO2:** NEGATIVE (-) — Healthy forests and soils absorb CO2 from the atmosphere through photosynthesis and storage, reducing atmospheric concentration. Deforestation or ecosystem degradation reverses this relationship.

Key Discoveries

- Even with immediate elimination of all fossil fuel emissions, Global Temperature Anomaly continues rising for decades due to thermal inertia in the ocean and activated feedback loops — the climate system has significant lag time
- Positive feedback loops like Ice-Albedo Feedback and Permafrost Methane Release can become self-sustaining once triggered, meaning they continue driving warming independently of human emissions
- Ocean Carbon Absorption decreases as ocean temperature and acidity increase, creating another positive feedback loop — warmer oceans absorb less CO2 which increases warming which further reduces absorption
- Combining emission reductions with enhanced carbon sinks produces the most favorable outcomes, but even the best scenarios show continued warming for 20-40 years before stabilization — there is no instant fix

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast

Full period available	Component Jigsaw — deep collaboration, every student contributes
After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Climate change will stop as soon as we stop emitting CO2"	Due to thermal inertia (the ocean has absorbed decades of excess heat) and activated feedback loops (ice-albedo, permafrost thaw), warming continues for 20-40 years after emissions cease. Some changes like ice sheet melting may continue for centuries. This does not mean action is pointless — stopping emissions prevents additional tipping points from being crossed — but it does mean there is a time lag between action and result.
"The ocean will keep absorbing our CO2 indefinitely"	The ocean\'s CO2 absorption capacity decreases with temperature and increasing acidity. Warmer water dissolves less gas (like how warm soda goes flat), and acidification from absorbed CO2 is already disrupting marine chemistry. The ocean currently absorbs about 25% of emissions, but models project this fraction will decline as warming continues, turning a brake on warming into a weaker and weaker buffer.
"Individual actions like recycling can solve climate change"	While individual actions matter for building cultural momentum, the scale of the climate challenge is systemic. Individual consumer choices account for a small fraction of global emissions compared to industrial, energy, and transportation systems. Solving climate change requires systemic policy changes — carbon pricing, renewable energy mandates, industrial regulation, and international cooperation. The model shows that only large-scale changes to Fossil Fuel Emissions and Terrestrial Carbon Storage produce meaningful changes in temperature trajectories.